NeurIPS 2026 Workshop Paper Guidelines

STORM-World: Physics-Informed, Intervention-Aware World Modeling for Tropical Cyclones

Target workshop: World Models in Physical AI | NeurIPS 2026
Submission deadline: September 5, 2026, AoE | Maximum length: 8 pages excluding references

Recommended central message

A useful physical world model must do more than forecast accurately: it must preserve physical dynamics, represent uncertainty, respond coherently to interventions, and preserve meaningful action rankings.

Source work	Role in workshop paper
STORM-CARE (AAAI 2027)	Provides the physics-informed encoder/operator, heterogeneous disaster graph, RSSM latent world model, open-loop evaluation, and intervention simulator.
AOTS2Action (IEEE BigData 2026)	Provides uncertainty-preserving ensemble analytics, real geospatial exposure layers, calibration/sharpness evaluation, and regional ranking ideas.

Workshop website: https://www.worldmodels-physicalai.com/

1. Strategic Positioning

The workshop paper should not be positioned as another hurricane-forecasting paper or as a shortened version of STORM-CARE. Its central scientific question should be whether a learned physical world model can imagine alternative storm-disaster futures under interventions while maintaining physical plausibility and useful decision structure.

Core question: Can a learned world model simulate physically plausible storm-disaster futures and preserve meaningful intervention effects under alternative actions?

The conceptual shift is:

Conventional framing	Workshop framing
Forecast the future	Imagine alternative futures under actions/interventions
Optimize prediction accuracy	Evaluate prediction + physics + uncertainty + intervention fidelity
Storm track/intensity as outputs	Coupled world state as a controllable latent dynamical system
Humanitarian outputs as endpoints	Decision consequences of imagined worlds

Recommended title

Preferred: STORM-World: A Physics-Informed Intervention-Aware World Model for Counterfactual Storm Simulation

Alternative: Learning Physically Grounded World Models for Intervention-Aware Storm Simulation

2. How This Paper Must Differ from Existing STORM Work

Paper	Main question	Primary contribution
IEEE BigData - AOTS2Action	How should ensemble uncertainty propagate into downstream exposure?	Uncertainty-preserving geospatial analytics
AAAI - STORM-CARE	Can physics-informed pretrained models forecast coupled storm/disaster states?	Forecasting + physics + humanitarian prediction + world-model framework
NeurIPS Workshop - STORM-World	Can a world model imagine physically consistent alternative futures under interventions?	World dynamics + controllability + intervention fidelity + decision-oriented evaluation

The workshop manuscript should therefore concentrate on what happens after a world state has been learned: open-loop rollout, intervention branching, controllability, physical fidelity, uncertainty, and decision preservation.

3. World-Model Formulation

Recast the current STORM-CARE architecture as an explicit action-conditioned world model.

```
World state:  W_t = {H_t, I_t, P_t, R_t}
```

- **H_t:** storm/hazard state
- **I_t:** infrastructure state
- **P_t:** population/exposure state
- **R_t:** response/resource state

```
Latent encoding:  z_t = E(W_t)

Transition:  z_(t+1) ~ p_theta(z_(t+1) | z_t, a_t, c_t)
```

- **z_t:** latent physical/disaster world state
- **a_t:** human intervention or action
- **c_t:** exogenous forcing, such as storm intensity or external system failure

```
Decoded future:  W_hat_(t+1) = D(z_(t+1))

Rollout:  tau_hat^(a) = {W_hat_(t+1)^(a), ..., W_hat_(t+H)^(a)}
```

The paper should compare a factual/baseline rollout against multiple intervention-conditioned rollouts originating from the same observed warm-up state.

4. Be Precise About the Term "Counterfactual"

Do not claim that the current system identifies real-world causal effects of evacuation or other interventions. The current evidence supports model-generated intervention scenarios, not causal identification from observational data.

Recommended wording: "We use counterfactual to denote alternative model-generated intervention scenarios rather than identified real-world causal effects."

Use terms such as "intervention-conditioned rollout," "alternative scenario simulation," or "model-internal counterfactual rollout." Avoid statements such as "the model estimates the causal effect of evacuation" unless a true causal identification design is added.

5. Primary Intervention Story: Evacuation Timing

Evacuation should be the flagship intervention because it is the strongest and most statistically stable result in the current STORM-CARE experiments.

Scenario	Mean peak exposure	Interpretation
Delayed evacuation	0.2969	Higher exposure than baseline
Baseline	0.2915	Reference
Evacuation 12 h earlier	0.2831	Lower exposure
Evacuation 24 h earlier	0.2787	Further reduction
Evacuation 36 h earlier	0.2754	Lowest exposure

The ordering -36 h < -24 h < -12 h < baseline < delayed is especially valuable as a world-model controllability result. The current analysis reports monotone ordering in all 10,000 bootstrap resamples, and the adjacent reductions remain significant after Holm correction.

Reframe this result as a controllability/dose-response test: does a stronger or earlier action produce an ordered, stable change in the imagined future?

6. Do Not Hide Failed Intervention Channels

The model currently does not respond reliably to every intervention. This should become an explicit evaluation contribution rather than being hidden as an inconvenient limitation.

Intervention	Expected direction	Current learned response	Assessment
Earlier evacuation	Exposure decreases	Decreases monotonically	Supported
Delayed evacuation	Exposure increases	Increases	Supported
Storm intensity increase	Hazard/risk increases	Correct sign, weak magnitude	Partial
Storm intensity decrease	Hazard/risk decreases	Correct sign, weak magnitude	Partial
Road blockage	Risk/exposure increases	Near-null / wrong sign	Failure
Shelter capacity failure	Risk/exposure increases	Wrong sign	Failure
Hospital service failure	Risk/exposure increases	Wrong sign	Failure
Additional resources	Risk/exposure decreases	Approximately null	Failure

This motivates a workshop-relevant question: when is a world model intervention-faithful rather than merely predictively accurate? The failure analysis can be scientifically valuable because the workshop explicitly focuses on physical correctness, causal fidelity, controllability, and downstream decision utility.

7. Recommended Research Questions

RQ1 - Predictive World Fidelity: Can the latent world model maintain accurate storm/disaster-state rollouts as the horizon increases from 6 to 120 hours?

RQ2 - Physical Fidelity: Does physics-informed learning reduce physically implausible trajectories and long-horizon drift?

RQ3 - Intervention Controllability: Does modifying an intervention produce coherent and directionally meaningful changes in imagined futures?

RQ4 - Intervention Fidelity and Failure: Which interventions are faithfully represented, and when does the learned world model fail to preserve expected intervention relationships?

8. New Metric: Intervention Consistency

Add a simple world-model-specific metric that measures whether an intervention changes the predicted future in the expected direction.

```
C(a) = (1/N) * sum_i  1[ sign(Delta_i^a) = s_a ]
```

Here s_a is the expected direction of change. Examples: earlier evacuation should reduce exposure; delayed evacuation should increase exposure; increased storm intensity should increase hazard. Report an Intervention Consistency Rate for each action and an overall macro-average across actions.

9. Stronger New Experiment: Policy-Ranking Preservation

A useful decision world model may not require perfect state prediction if it still ranks interventions correctly. This should be one of the most workshop-specific experiments.

```
a* = argmin_a  R(tau_hat^(a))
```

For the evacuation family, test whether the world model preserves the ranking:

```
Evac(-36 h) > Evac(-24 h) > Evac(-12 h) > Baseline > Delayed
```

Evaluate with:

- Spearman rank correlation between predicted and reference intervention orderings.
- Kendall tau for pairwise ranking consistency.
- Top-1 action selection accuracy.
- Pairwise intervention-ranking accuracy.
- Decision regret relative to the best intervention under the reference simulator or verified target.

10. Connect AOTS2Action to STORM-World

If feasible before the deadline, use AOTS2Action to strengthen the downstream evaluation with real geospatial data. AOTS2Action already provides uncertainty-preserving forecast ensembles, WorldPop exposure, socioeconomic vulnerability, administrative boundaries, IBTrACS verification, and regional prioritization.

```
World-model rollout -> ensemble future states -> q_h(x) -> exposure ->
regional priority
```

This integration would reduce dependence on purely synthetic humanitarian targets and allow a more realistic test of whether intervention-conditioned world states lead to useful downstream spatial risk estimates.

Important: keep the novelty centered on the learned world model and intervention-conditioned imagination. Do not turn the workshop paper into AOTS2Action II.

11. Preserve Uncertainty Through the World Model

AOTS2Action demonstrates the value of not collapsing multiple plausible trajectories too early. Apply the same principle to STORM-World: model a distribution over future worlds rather than only one deterministic rollout.

```
p_theta(W_(t+1:t+H) | W_t, a)
```

For each intervention, generate multiple imagined futures and estimate quantities such as P(exposure | a), interval coverage, trajectory diversity, and action-ranking stability under sampled futures. Compare stochastic versus deterministic rollout variants.

12. Required Ablations

Variant	Question answered
Full STORM-World	Main method
- Physics regularization	Does physical grounding improve long-horizon plausibility?
- Pretraining	Does pretrained representation improve world-state fidelity?
- World model / direct predictor	Is latent multi-step rollout actually useful?
- Intervention conditioning	Does explicit action information matter?
Deterministic rollout	What is gained by stochastic imagined futures?
Optional: - Graph structure	How much does structured interaction modeling matter?

13. Evaluation Framework: Four Dimensions

Dimension	Recommended metrics
Predictive fidelity	Track error, intensity MAE, state reconstruction error, exposure error, multi-horizon rollout degradation
Physical fidelity	Advection, diffusion, mass consistency, wind-pressure consistency, temporal continuity, energy consistency
Probabilistic fidelity	P50/P90 coverage, calibration error, sharpness, Brier score, stochastic rollout diversity
Intervention/decision fidelity	Intervention consistency, dose-response, action-ranking correlation, top-1 action accuracy, decision regret

The current uncertainty results already provide a useful long-horizon failure story: calibration is much better at short horizons and degrades substantially at longer horizons. Present this as world-model uncertainty drift rather than as a side calibration result.

14. Main Architecture Figure

Do not reuse a complicated end-to-end architecture figure unchanged. The workshop figure should immediately communicate one observed world branching into multiple intervention-conditioned imagined worlds.

Observed physical world -> Physics-informed encoder -> Latent world state z_t -> Action/intervention branches -> Open-loop world rollouts -> Future risk / decision comparison

Suggested branches: No action, Evacuate 12 h earlier, Evacuate 24 h earlier, Evacuate 36 h earlier. A separate visual encoding can distinguish human actions from exogenous perturbations such as storm-intensity changes or infrastructure failures.

15. Recommended Figures and Tables

Item	Content
Figure 1	STORM-World architecture: one observed world branching into action-conditioned latent rollouts
Figure 2	Alternative futures for one cyclone: baseline vs. -12/-24/-36 h evacuation
Figure 3	Physical consistency and predictive error versus rollout horizon, with and without physics
Figure 4	Intervention fidelity matrix showing supported, partial, and failed channels
Table 1	Predictive + physical fidelity: full model and ablations
Table 2	Evacuation-conditioned rollout results and ranking statistics
Table 3	All intervention channels with expected sign, learned sign, effect magnitude, and consistency

16. Three Contributions Only

- **Contribution 1 - Physically grounded storm world model.** Formulate coupled storm, infrastructure, population, and response dynamics as a structured latent world model capable of open-loop multi-step simulation.
- **Contribution 2 - Intervention-conditioned imagination.** Branch a common observed state into alternative future trajectories under evacuation and environmental/system perturbations.
- **Contribution 3 - Evaluation beyond forecasting accuracy.** Evaluate predictive fidelity, physical consistency, uncertainty calibration, intervention consistency, decision ranking, and explicit failure modes.

17. Recommended 8-Page Organization

Section	Approx. space	Purpose
1. Introduction	0.8 page	World-model problem; intervention fidelity; contributions
2. Related Work	0.6 page	Weather foundation models, physical world models, latent dynamics, action-conditioned models
3. Problem Formulation	0.6 page	State, latent state, action, rollout, evaluation target
4. STORM-World	1.6 pages	Physics-informed encoder/operator, RSSM, intervention branching, stochastic rollout
5. Experimental Design	1.0 page	Data, splits, interventions, baselines, metrics
6. Results	2.1 pages	Predictive, physical, uncertainty, intervention and failure results
7. Discussion & Limitations	0.7 page	What makes a world model physically and intervention-faithful?
8. Conclusion	0.3 page	Takeaway and broader physical-AI implication

18. Recommended Opening of the Paper

> *"World models are increasingly expected to do more than predict the next state of a physical environment: they must support reasoning about how the environment would evolve under alternative actions. Yet predictive accuracy alone does not establish that an imagined future is physically plausible or intervention-faithful. Tropical cyclones provide a challenging testbed for this problem because atmospheric dynamics interact with infrastructure, population exposure, and human interventions over long horizons."*

Then introduce STORM-World immediately as a physics-informed, intervention-aware latent world model for simulating alternative storm-impact futures.

19. What to Reuse from STORM-CARE

Reuse prominently	Do not make central
Frozen storm-level splits; physics-informed graph-Fourier operator; pretrained encoder; RSSM; open-loop evaluation; physics ablations; uncertainty calibration; evacuation dose-response; Hurricane Ian visualization; intervention failure cases.	Five humanitarian prediction heads; lengthy RF/XGBoost comparisons; extensive school/hospital output discussion; broad general forecasting benchmark; every pretraining objective.

The existing temporal split is already a strong foundation: 342 training storms (1995-2015), 70 validation storms (2016-2019), and 107 test storms (2020-2024), with no storm appearing in more than one partition.

20. What to Reuse from AOTS2Action

- Member-level ensemble preservation rather than immediate collapse to a mean track.
- Probability-weighted exposure based on ensemble support.
- WorldPop and real administrative geospatial layers.

- IBTrACS best-track verification.
- Coverage and sharpness evaluation.
- Brier score for probabilistic exposure fidelity.
- Regional ranking metrics such as nDCG and ranking correlation.

Use these elements to strengthen world-model evaluation, but maintain the central novelty as learned intervention-conditioned physical dynamics.

21. Priority Work Plan Before September 5

Priority	Task	Purpose
P0	Reframe the RSSM as an explicit action-conditioned world model	Essential conceptual alignment
P0	Create a new STORM-World architecture figure	Make workshop fit visually obvious
P0	Run/report evacuation action-ranking experiment	Strongest new decision-oriented result
P0	Create intervention-direction/fidelity table	Turn failure cases into a contribution
P0	Plot physics consistency versus rollout horizon	Connect physical grounding to long-horizon world modeling
P0	Include failed intervention channels explicitly	Avoid overclaiming causal/world-model maturity
P1	Integrate AOTS real-geospatial exposure with rollouts	Strengthen downstream realism
P1	Compare deterministic vs. stochastic futures	Demonstrate uncertainty-aware world modeling
P1	Report calibration versus rollout horizon	Measure long-horizon uncertainty drift
P2	Increase world-model training scale	Potentially improve weak interventions
P2	Repair route/facility response failures	Only if time permits
P2	Multi-basin evaluation	Future extension if not feasible now

22. Critical Guidance to the Team

Do not spend the remaining time trying to make every intervention look successful. The current demo-scale world model already provides a more interesting scientific story: a model may forecast reasonably and satisfy several physical constraints while still failing to respond faithfully to some interventions.

Potential workshop thesis: Predictive accuracy is not enough. Physical world models should be evaluated for physical consistency, uncertainty calibration, intervention controllability, and decision-ranking fidelity - including explicit failure modes.

This framing makes the paper a contribution to world-model evaluation rather than only an application of AI to tropical cyclones.

23. Final Paper Message

A useful physical world model must satisfy more than forecast accuracy. It must preserve physical dynamics, represent uncertainty, remain controllable under interventions, and preserve meaningful action rankings. STORM-World uses tropical cyclones as a real physical-AI testbed for studying these requirements.

24. Working Source Materials

- STORM-CARE: A Physics-Informed Pretrained World Model for Disaster-State Forecasting and Counterfactual Intervention Simulation (current AAAI 2027 manuscript/supplement).
- AOTS2Action: Uncertainty-Aware Big Data Analytics for Humanitarian Cyclone Risk (current IEEE BigData 2026 manuscript).

- World Models in Physical AI, NeurIPS 2026 workshop (official site): https://www.worldmodels-physicalai.com/

25. Dataset and External Data URLs

Use the exact dataset snapshot, version, date range, and checksum used in each experiment. Do not silently replace the frozen STORM-CARE data with a newer public release. The URLs below are the authoritative or project-relevant access points to cite in the workshop manuscript and repository.

Dataset / resource	Recommended role in STORM-World	Official / primary URL
HURDAT2 - NHC best-track data	Core STORM-CARE storm track/intensity source. The public NHC page now serves updated releases; preserve the frozen project snapshot for reproducibility.	https://www.nhc.noaa.gov/data/
ERA5 pressure-level reanalysis	Atmospheric/reanalysis variables for physics-informed forecasting and stronger future experiments. Report exact variables, levels, resolution, coverage, and acquisition date.	https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels
CDC/ATSDR Social Vulnerability Index (SVI)	STORM-CARE vulnerability features. Use the documented SVI vintage actually consumed by the frozen run.	https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html
IBTrACS v4r01	AOTS2Action best-track verification and recommended multi-basin tropical-cyclone validation source. The frozen STORM-CARE configuration did not consume IBTrACS, so do not imply that it did.	https://www.ncei.noaa.gov/products/international-best-track-archive
WorldPop 2020 population counts	AOTS2Action real population layer; useful for converting world-model rollouts into probability-weighted exposure.	https://hub.worldpop.org/geodata/summary?id=24777
Natural Earth	AOTS2Action administrative boundaries and contextual airports/ports; also suitable for publication maps.	https://www.naturalearthdata.com/downloads/
World Bank country and lending groups	Country-level income classification used as the socioeconomic vulnerability proxy in AOTS2Action. Freeze the classification vintage used in the experiment.	https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups
TIGGE ensemble forecasts - ECMWF	Recommended external multi-model ensemble source for uncertainty-aware validation or future replacement/augmentation of internal ensemble data.	https://ecds.ecmwf.int/datasets/tigge-forecasts?tab=overview
ECMWF Open Data	Optional current operational IFS/AIFS forecast source for a small contemporary case study; use only if acquisition and verification protocol can be frozen before submission.	https://www.ecmwf.int/en/forecasts/datasets/open-data

Data-version warning: The current STORM-CARE supplement reports very limited ERA5 coverage in the frozen repository and a frozen HURDAT2-based split. Any expanded ERA5 or IBTrACS experiment for the workshop must be labeled as a new experiment rather than silently merged into the old benchmark.

26. Baseline Models and Implementation URLs

Separate baselines into (A) required executable comparisons that can be reproduced before September 5 and (B) strong external weather-model references that should be executed only if inputs, compute, and evaluation are genuinely comparable. If a model is cited but not run, label it as related work rather than a numerical baseline.

Baseline	Priority	Why it matters	URL
Persistence / extrapolation	Required simple baseline	WeatherBench-X includes persistence loaders; for TC-specific practice also follow NHC verification conventions.	https://weatherbench-x.readthedocs.io/en/latest/
CLIPER5	Required TC skill baseline where feasible	NHC climatology-and-persistence track model used to measure forecast skill.	https://www.nhc.noaa.gov/modelsummary.shtml
LSTM	Required sequence baseline	Standard recurrent baseline for temporal dynamics.	https://doi.org/10.1162/neco.1997.9.8.1735
Transformer	Required sequence baseline	Standard attention-based temporal baseline.	https://arxiv.org/abs/1706.03762
DCRNN	Required graph-temporal baseline	Diffusion-convolution recurrent model; useful as a graph sequence comparator.	https://github.com/liyaguang/DCRNN
Fourier Neural Operator / NeuralOperator	Required operator-family reference	Comparator for learned PDE/operator dynamics and the graph-Fourier design.	https://arxiv.org/abs/2010.08895
FourCastNet	Strong external weather baseline	High-resolution data-driven global weather forecasting using adaptive Fourier neural operators.	https://github.com/NVlabs/FourCastNet
Pangu-Weather	Strong external weather baseline	3D neural global weather model; includes public inference resources.	https://github.com/198808xc/Pangu-Weather
GraphCast	Strong external weather baseline	Graph-based global medium-range weather model; code/weights are public.	https://github.com/google-deepmind/graphcast
GenCast	Strong probabilistic weather baseline	Stochastic ensemble weather model; example code is distributed in the same DeepMind repository.	https://github.com/google-deepmind/graphcast
NeuralGCM	Strong hybrid physics/ML baseline	Differentiable hybrid general circulation model relevant to physical consistency.	https://github.com/neuralgcm/neuralgcm
Aurora	Strong foundation-model baseline	Earth-system foundation model with tropical-cyclone forecasting capabilities.	https://github.com/microsoft/aurora
ECMWF IFS / ENS	Operational reference	Use WeatherBench/TIGGE or ECMWF open data when the comparison protocol is truly like-for-like.	https://www.ecmwf.int/en/forecasts/datasets/open-data

Minimum executable baseline set for this workshop: Persistence, CLIPER5 when compatible with the track protocol, LSTM, Transformer, DCRNN or another graph-temporal model, full STORM-World, -physics, -pretraining, -world-model/direct prediction, and deterministic-vs-stochastic rollout. GraphCast/Pangu/NeuralGCM/GenCast/Aurora are valuable only if a fair input-aligned evaluation can be completed; otherwise discuss them in Related Work.

27. Benchmarking Resources, Protocols, and URLs

The workshop paper should benchmark four distinct properties: forecast fidelity, physical fidelity, probabilistic calibration, and intervention/decision fidelity. Use established weather-verification resources for the first three, while clearly identifying the intervention metrics as STORM-World-specific additions.

Benchmark / protocol	Recommended use	URL
WeatherBench 2 benchmark	Like-for-like global deterministic/probabilistic weather evaluation and public scorecards.	https://sites.research.google/gr/weatherbench/
WeatherBench 2 data guide	Cloud-optimized ERA5, HRES/ENS and other benchmark datasets; useful for aligned external comparisons.	https://weatherbench2.readthedocs.io/en/latest/data-guide.html
WeatherBench-X	Current modular successor to the WeatherBench 2 evaluation code; supports deterministic, probabilistic, spatial and sparse-observation evaluation.	https://weatherbench-x.readthedocs.io/en/latest/
WeatherBench-X code	Implementation reference if the team adopts its loaders/metrics or adapts it for storm-specific evaluation.	https://github.com/google-research/weatherbenchX
NHC Forecast Verification	Authoritative TC track/intensity verification conventions, historical errors, skill measures and annual reports.	https://www.nhc.noaa.gov/verification/
NHC verification procedures	Defines best-track-based track and intensity errors and operational verification lead times.	https://www.nhc.noaa.gov/verification/verify2.shtml
NHC official forecast error database	Historical track/intensity forecast errors; useful for contextualizing TC forecast skill.	https://www.nhc.noaa.gov/verification/verify7.shtml
TIGGE	Multi-centre ensemble forecasts for probabilistic/ensemble benchmarking and tropical-cyclone case studies.	https://ecds.ecmwf.int/datasets/tigge-forecasts?tab=overview

Recommended benchmark scorecard for STORM-World

Evaluation dimension	Recommended measures
Forecast fidelity	Great-circle track error; intensity MAE; state RMSE; horizon-specific errors at 6/12/24/48/72/96/120 h as available.
Physical fidelity	Advection, diffusion, mass/divergence, wind-pressure, continuity, and energy residuals; plot residual growth with rollout horizon.
Probabilistic fidelity	P50/P90 coverage, calibration error, sharpness/area, Brier score for exposure fields, CRPS where full predictive distributions are available, spread-skill ratio where appropriate.
Intervention fidelity	Intervention Consistency Rate; dose-response monotonicity; sign correctness; effect magnitude sensitivity; failure-channel audit.
Decision fidelity	Pairwise action-ranking accuracy, Spearman/Kendall ranking, top-1 action accuracy, and decision regret.
Statistical protocol	Resample at storm level for storm forecasts and sequence level for world-model interventions; do not treat correlated forecast windows as independent samples.

28. Related Work Map with Paper URLs

Keep Related Work short and organized around the paper's methodological question. The most useful references are those that define world models, physical inductive bias/operator learning, modern AI weather forecasting, probabilistic ensembles, and Earth-system foundation models.

Theme	Key reference / relevance	URL
World models	Ha & Schmidhuber, "World Models" - compressed latent dynamics and imagined rollouts.	https://arxiv.org/abs/1803.10122

Theme	Key reference / relevance	URL
Action-conditioned world models	Hafner et al., DreamerV3 - learning and acting from imagined trajectories.	https://www.nature.com/articles/s41586-025-08744-2
Physics-informed learning	Raissi, Perdikaris & Karniadakis, Physics-Informed Neural Networks.	https://doi.org/10.1016/j.jcp.2018.10.045
Neural operators	Li et al., Fourier Neural Operator for parametric PDEs.	https://arxiv.org/abs/2010.08895
AI weather forecasting	Pathak et al./Kurth et al., FourCastNet.	https://arxiv.org/abs/2202.11214
AI weather forecasting	Bi et al., Pangu-Weather, Nature 2023.	https://doi.org/10.1038/s41586-023-06185-3
Graph weather models	Lam et al., GraphCast, Science 2023.	https://doi.org/10.1126/science.adi2336
Hybrid physics + ML	Kochkov et al., NeuralGCM, Nature 2024.	https://doi.org/10.1038/s41586-024-07744-y
Probabilistic weather world modeling	Price et al., GenCast, Nature 2024 - stochastic ensemble trajectories and calibration.	https://doi.org/10.1038/s41586-024-08252-9
Earth-system foundation models	Bodnar et al., Aurora, Nature 2025 - foundation model across weather/Earth-system tasks including TC tracks.	https://doi.org/10.1038/s41586-025-09005-y
Weather benchmark	Rasp et al./WeatherBench 2 ecosystem - standardized evaluation and public baselines.	https://sites.research.google/gr/weatherbench/
TC forecast skill baselines	NHC guidance suite / CLIPER5 and operational model guidance.	https://www.nhc.noaa.gov/modelsummary.shtml

Recommended positioning sentence: Prior AI weather models primarily optimize forecast skill or probabilistic ensemble quality, while general world-model work emphasizes latent action-conditioned imagination. STORM-World should position itself at their intersection: a physics-informed storm-disaster world model evaluated not only for forecast and calibration quality, but also for whether interventions produce directionally coherent futures and preserve decision rankings.

URL verification note: Public links in Sections 25-28 were checked against official dataset/project/publisher pages on August 29, 2026. For the manuscript, cite the paper/DOI and dataset version in the bibliography; use raw URLs mainly in the project README, reproducibility appendix, or data/code availability statement.